Chapter 7 – PKI and Cryptographic Applications

Asymmetric Cryptography

- We talked about Symmetric lets talk about Asymmetric

Asymmetric Cryptography

- Public and Private Keys
- RSA (Rivest-Shamir-Adleman)
  - Standard today
  - Widely use for secure communication
  - Relies on computational difficulty

ElGamal

- Diffie Hellman uses large integers
- Major advantages over RSA was it was released to the public
- The major disadvantage doubles the size of the message it encrypts

Elliptic Curve Cryptography (ECC)

- Elliptic Curve Discrete Logarithm Problem
- Extremely difficult to decrypt
- 3072 bit RSA = 256 bit elliptical curve cryptosytem

Diffie-Hellman Key Exchange

- In Chapter 6 we learned that we can generate a shared key over insecure channels
- This can be used to create a public key as a hybrid cryptosystem
- TLS (Transport Layer Security)

Quantum Cryptography

- Theoretical research
- No useful practical implementation of “quantum computer”
- Most of this relies on potential

Quantum Cryptography

- There are lab implementation of quantum key distribution (QKD)
- Post-Quantum Crpytography

Hash Function

- Verify files
- Main purpose: They take a potentially long message and generate a unique output value derived from the content of the message
- If hashes don’t match, someone altered it
- One-Way Function

SHA Family

- SHA-1, SHA-2, SHA-3
- SHA-1 is no longer recommended for any purpose
- Replaced with SHA-2 that has four major variants

SHA Family

- SHA-256 produces a 256-bit message digest using a 512-bit block size
- SHA-224 uses a truncated version of the SHA-256 produces 224-bit message digest using a 512-bit block size

SHA Family

- SHA-512 produces a 512-bit message using 1024-bit block size
- SHA-384 truncated version of SHA-512 produces a 384-bit digest using a 1024-bit block size

SHA Family

- In 2015, SHA-3 was developed to replace SHA-2
- SHA-3 isn’t used as it gives the same security as SHA-2 but slower

MD5

- Historical but still used
- Main vulnerability is collision
- It is much better to rely on the SHA family

RIPEMD

- Alternative to the SHA family of hashing
- RIPEMD-128 (no longer secured)
- RIPEMD-160 (used today)
- RIPEMD-256 (equivalent security with 128)
- RIPEMD-320 (equivalent security with 320)

Digital Signatures

- Once you chose a hash function and public key algorithm you can make a digital signature system now
- There are two goals

Digital Signatures

- Assure the recipient that message truly came from the claimed sender
- Assure the recipient that the message was not altered while in transit between the sender and recipient

Public Key Infrastructure (PKI)

- Digital certificates
- Certificate Authorities
- Certificate Life Cycle
  - Enrollment
  - Verification
  - Revocation

Certificate Formats

- .der
- .crt
- .cer
- .pem
- .pfx
- .p12

Asymmetric Key Management

- We learned “security through obscurity” (not appropriate)
- Retire keys when they’ve served a useful life
- Back up your key
- Hardware security modules (HSMs)

Hybrid Cryptography

- Using both symmetric and asymmetric cryptosystem
- The first connection uses asymmetric, then uses symmetric
- Benefits of security of asymmetric then enjoying the speed of symmetric e.g. TLS

Applied Cryptography

- Portable Devices
- Trusted Platform Module (TPM)
- Email
- Pretty Good Privacy (PGP)
- S/MIME

Applied Cryptography

- Web Application
- Secure Sockets Layer (SSL)
- Transport Layer Security (TLS)
- Tor and the Dark Web
- Steganography and Watermarking

Applied Cryptography

- Networking
- Circuit Encryption
  - Link encryption
  - End-to-End encryption
- IPSec

Emerging Applications

- Blockchain
- Lightweight Cryptography
- Homomorphic Encryption

Cryptographic Attacks

- Brute-Force Attacks
- Analytic Attacks
- Implementation Attacks
- Statistical Attacks

Cryptographic Attacks

- Fault Injection Attacks
- Side-Channel Attacks
- Timing Attakcs
- Frequency Analysis and Cipher Attacks

Cryptographic Attacks

- Man-in-the-Middle Attack
- Birthday Attack
- Replay Attack